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LESZEK MIS PROFILE / DEFENSIVE SECURITY OFFER PACKAGE

An offer from Defensive Security Holding Sp. z o.o., focused on Linux environment security from an offensive, detection, and forensics perspective. The presenter and author of the solutions described is Leszek Miś, owner of the company, but above all, an active Security Researcher, consultant, and trainer.

I would like to present the following components of the offer, which is also a broad description of my experience profile gained during over 20 years of adventure with Cyber Security:

1. PurpleLabs Cyber Range training environment.
2. EDRmetry Linux Matrix - Comprehensive Hands-On Attack TTPs Catalog.
3. Linux Attack, Detection and Forensics v2.0 - Hands-on Purple Teaming Playbook.
4. Purple Teaming Exercises / Detection Engineering.
5. Consultancy services and potential ways for collaboration.
6. About Defensive Security.
7. Leszek Mis Profile / Experience.

I. PURPLELABS CYBER RANGE:

PurpleLABS is a dedicated virtual infrastructure for conducting detection and behavioral analysis of attackers with respect to the offensive techniques, tactics, procedures, and tools they employ. The environment is designed to continuously raise competencies in threat hunting and to build familiarity with current offensive trends (red teaming) through a direct detection lens (blue teaming/forensics). By providing high-quality training materials together with a laboratory environment in a scalable online format, we aim to enable organizations to improve the detection capabilities of their security teams/SOC and achieve better visibility and resilience against attacks. Active participation of an organization's security team in the PurpleLabs program will allow them to:

- **Master Linux Attack Paths and Defenses:**
 - Get to know the newest Linux attack paths, hiding techniques, and offensive tools for Discovery, C2, Lateral Movement, Persistence, Evasion, Exfiltration, Execution, and Credential Access, aligned with the Linux MITRE ATT&CK Framework.
- **Enhance Detection and Correlation Skills:**
 - Learn ways to improve detection, sharpen event correlation across Linux/network data sources, and identify malicious activities to filter out network noise for better incident response.
- **Explore Detection and DFIR Tools:**
 - Understand visibility/detection methods of tools like Splunk, Elastic Security, Velociraptor, Falco, Jibril, Tracee, Tetragon, Kunai, Moloch/Arkime, Zeek, Suricata, OSquery, Wazuh, and Sandfly Security, including how open-source software supports SOC infrastructure.
- **Proactive Forensics and Emulation:**
 - Gain insights into proactive Linux forensics scans, manual/automated attacker simulations to generate anomalies, and identifying blind spots in network security posture.
- **Purple Teaming Approach:**

- Understand the value of purple teaming, where red and blue teams collaborate to hunt for threats, augment SOC efficiencies, and extend Breach and Attack Simulation Systems (BAS enrichment).
- **Offensive Techniques and Evasion:**
 - Learn full-scope Linux offensive techniques, tools, community research, stealthy hiding/exfiltration methods, C2 deployment, low-level rootkits, and defense evasion/persistence skills.
- **Linux Internals and Kernel Security:**
 - Recognize security enhancements in modern Linux kernels, understand kernel components/programming interfaces for system compromise, and gain experience with internals for better attack logic and telemetry understanding.

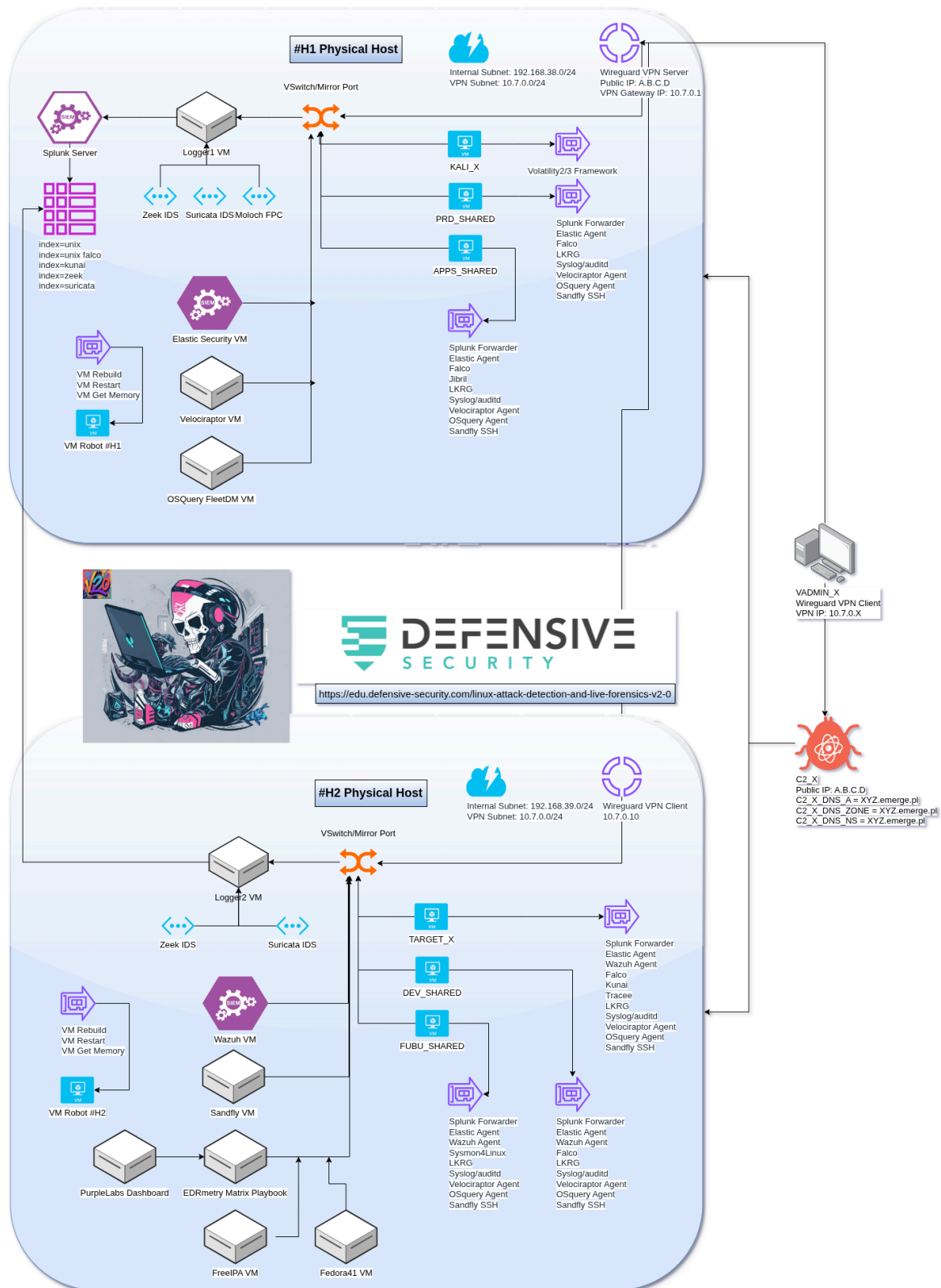
The PurpleLabs environment is modular, allowing new components to be added with ease. There is potential to launch a dedicated instance of the PurpleLabs environment within a "security testing" type environment inside the customer's network infrastructure.

As of today, the following system and network components are supported:

- **HOST:**
 - Syslog, Auditd, Elastic EDR, Falco Runtime Security, Kunai Runtime Security, Jibril Runtime Security, Tetragon Runtime Security, Tracee Runtime Security, Sysmon4Linux, Velociraptor IR, OSQuery/Sunlight, Sandfly Security, Linux IR Scripts, UAC, Yara, LKRG, SELinux, and more
- **NETWORK:**
 - Zeek NIDS, Suricata NIDS, Arkime/Moloch FPC, Modsecurity WAF
- **SIEM:**
 - Elastic, Splunk, Wazuh
- **MEMORY:**
 - Volatility2/3 Framework

- Automated RAM acquisition

The diagram illustrating the environment architecture is shown below:



II. EDRMETRY LINUX MATRIX

EDRmetry Matrix is a user-friendly central knowledge base on offensive techniques in the Linux environment, delivering ready-to-run, atomic execution instructions containing code snippets, scripts, exploits, and tools. Active use of EDRmetry Matrix enables cybersecurity professionals to efficiently familiarize themselves with the Linux threat landscape and to test the quality of collected telemetry, assess detection capabilities, tune EDR/SIEM stack configurations, and refine incident response procedures.

EDRmetry Matrix is the only technique emulation and attack chain platform available on the market dedicated exclusively to the Linux environment, featuring advanced and low-level offensive tests. The EDRmetry Linux Matrix offensive catalog contains definitions of over 450 offensive techniques (TTPs) derived from real-world attack scenarios within the Linux/Open Source environment. The EDRmetry platform thus delivers significant time savings for SOC/CSIRT teams while simultaneously enabling active evaluation and testing of visibility and detection, as well as building technical and low-level competencies through the approach of active defense - a proactive strategy aimed at preventing, detecting, and neutralizing threats in real time, rather than merely responding to attacks reactively.

matrices / EDRmetry Linux Matrix - Offensive TTPs Hands-on Catalog									
Command and Control 21 techniques	Credential Access 22 techniques	Defense Evasion 117 techniques	Discovery 34 techniques	Execution 15 techniques	Exfiltration 18 techniques	Impact 6 techniques	Initial Access 22 techniques	Lateral Movement 29 techniques	Persistence 83 techniques
C2 Implants (13)	Dump credentials via unshadow	ASM Injection over /proc/PID/mem	/proc/PID/ Enumeration	Built-in System Tools Execution	DNS Exfiltration with dig	Bash Fork Bomb	ActiveMQ CVE-2023-46604 Exploitation	Chisel Reverse Socks Proxy	/etc/modules-load.d Persistence
DNS AXFR Payload Delivery	Dump heap memory from Java	Avoid Filename and Filepath Matching	Access to the AWS Metadata Service	Dump process memory via GDB	DNS Tunneling/Exfiltration with dnscat2	Clear kernel ring buffer	Apache HTTP CVE-2021-41779 Exploitation	Cloudflared Tunneling	/etc/sudoers Modification
DNS Tunneling with Iodine	Dump Passwords From Proc Memory with mimipenguin	Base64 Payload as a filename inside ZIP	C2 randomized hostname lookups	eBPF system("whoami") Execution with bptrace	eBPF Magic String Tracpoint Execution with bptrace	Crypto Mining CPU stress	Apache Tomcat Manager Exploitation	Create a SOCKS proxy with ssh	Add Backdoor User - /etc/passwd modification
eBPF Keylogger + DNS RCE	Dump PostgreSQL data - pgread	Bash Anti-Forensic Log Wipe	Check ASLR configuration	Execute binary listening from a hidden directory as root	Exfil data using PUT method - transfer.sh	Ransomware bash+openssl	Apache Tomcat Spring4Shell CVE-2022-22965 Exploitation	DarkFlare TCP over CDN Tunneling	Add backdoor user with uid=0
Emp3rtr HTTP2 AES Stager C2	eBPF Capture TLS/SSL functions with Qtap	Bash HTTP GET data with /dev/tcp	Check bpf settings from /proc	Execute LKM call_usermodehelper() on ICMP	Exfil data using rsync	Ransomware Black Basta	HTTPD CVE-2014-6271 Shellshock RCE	DNS Potentially Suspicious Requests	Add new group
Execute UPX Reverse SSH server	eBPF pamspy	Bash Script Obfuscation	Dismap Asset Discovery	Execute miknod/mikfio	Exfil data with AWS S3		JetBrains TeamCity CVE-2023-42793	DNS Zone Transfer	Add User to Privileged Group
Fileless Reverse shell with ssh	eBPF Sniff pam_get_authtok() with bptrace	Bashrc File Hiding with ls Alias	DNS Reconnaissance	Export proxy_http	ICMP Python Scapy Exfiltration		Juicy Shop SQL Injection	Execute Port Scanning	At job persistence
goDoh - DNS over HTTPS C2	eBPF Sniff PTY with bptrace	Binary Runtime Crypter in Bash	DNS TXT request	Install suspicious RPM package	ICMP_exfil + nping Exfiltration		K8S - kubeconfig file	Execute SSHD as a victim user	Backdooring Intrafms
Make Non-standard port HTTP/HTTPS connection	eBPF Sniff TLS/TLS Traffic	Block rayslogd logging	Download and launch LinEnum	NTP Data Exfiltration	PAM creds over HTTP Post		K8S - kubeconfig file	Get malicious samples from MalwareBazaar	Bash/Dash MSF Backdoor
Reverse DNS Tunnel Backdoor	eBPF Sniff SSH/TLS Traffic	BOF Loading with BOF-Stager	Enumerate kernel modules	PAM Data Exfiltration	Python FTP Upload		Kafka CVE-2023-25194 Exploitation	Hijacking SSH Client Session	B0B Trace Hooking Rootkit
Reverse shells (30)	Find local passwords/secrets	Bypass TLS Certificate Verification via LD_PRELOAD	Enumy ELF	PAM Data Exfiltration	Python FTP Upload		Langflow API Pre-Auth CVE-2025-3248 Exploitation	ioX - Encrypted SOCKS5 and Port Forwarding	b0v1 - Patch Dynamic Linker
Reverse SSH - SSRImpersonate	Get AWS EC2 Credentials from ConfigMaps	Change Shell Optional Behavior	Execute "What Server" Enumeration	K8S - Exec into pod	SMB Data Exfiltration with impactet		MySQL Brute Force	KCP - FRP Fast Reverse Proxy	BPFDoor Backdoor
SCTP Bind Shell	K8S - Access Credentials from ConfigMaps	Cleaning from /var/log/secure	Execute BeRoot	K8S - Sidcar Injection	Telegram Data Exfiltration		Next.js React2shell - CVE-2025-55182	Ligolo-ng Reverse TCP/TLS Tunneling	Cap_setuid over LD linker
Shell Over Reverse SSH	K8S - Enumerate cluster secrets	Cleaning journal logs	Execute nping	LKM Load/unload kernel module	Upload data over HTTP/HTTPS		Orbis CVE-2024-45507 SSRF+RCE	Logi-ng Reverse TCP/TLS Tunneling	Crontab root Backdoor
SOA/ECS DNS C2 Channel with siphondns	K8S - Dump etcd database	Clear kernel ring buffer	Find all suid/sgid files	Python - File download without curl	Upload data over WebDAV		OpenSMTPD CVE-2020-7247 RCE	Logi-ng Reverse TCP/TLS Tunneling	Custom SSHD id_ed25519 Key Backdoor
SSH-based Reverse Shell from NIKAS	K8S - Enumerate cluster secrets	Clear Paging Cache	Find Email Addresses in FS	Python GET File over Network	Upload data over WebDAV		Oracle WebLogic SSRF Exploitation	Logi-ng Reverse TCP/TLS Tunneling	eBPF Boopkit Rootkit
SSL/TLS Revoked Certificate Connection Attempt	K8S - Steel Pod Service Account Token	Copy/rename commands to exotic directory	Find loaded eBPF programs/maps	Renice or Ulimit Execution	Upload/download data over SSHFS		PhrackCTF vnd_server UAUF+Heap Overflow Exploitation	Logi-ng Reverse TCP/TLS Tunneling	Deploy Malicious Docker Container
SSL/TLS Self-signed Certificate Connection	LaZagne Creds Recovery	Create file with Unicode zero-width space	Find SSH keys	Updog HTTP/S Server			Redis CVE-2025-49844 RedisAuth Exploitation	Logi-ng Reverse TCP/TLS Tunneling	Deploy malicious RPM package
			K8S - Enumerate nodes					Logi-ng Reverse TCP/TLS Tunneling	DNF Package Manager
								Logi-ng Reverse TCP/TLS Tunneling	Netesex - Network Service Exploitation Tool
								Logi-ng Reverse TCP/TLS Tunneling	eBPF Boopkit Rootkit
								Logi-ng Reverse TCP/TLS Tunneling	eBPF BPFDoor Backdoor

EDRmetry Matrix serves as an integral, hands-on component of the training and workshops, enabling dynamic and comprehensive exploration of given offensive technique families. Through the built-in keyword-based technique search functionality, it is possible to efficiently pivot and chain individual techniques into offensive event sequences, while simultaneously building advanced detection paths.

EDRmetry Values:

- **Controlled Offensive Testing, Smarter Linux Defensive Outcomes:**
 - Chaining offensive techniques, allowing your team to validate the effectiveness of Linux EDR or Runtime Security engines in real-time. EDRmetry ensures your defenses aren't just theoretical - they're tested and proven.
- **See What Your SIEM Can't:**
 - Uncover blind spots in your detection pipelines, telemetry flows, and data source correlations. By replicating adversary behavior, EDRmetry highlights exactly where your current tools are falling short, so you can close the gap before attackers exploit it.
- **Elevate Your Incident Response:**
 - Use EDRmetry as a foundation for meaningful internal Red vs Blue team exercises. Improve coordination, sharpen your response strategy, and gain deeper insight into Linux-specific TTPs and forensic artifacts
- **Enhance Detection Engineering and Threat Hunting:**
 - Focus your resources where they matter most by building better detections and expanding your threat-hunting capabilities. With EDRmetry, you maintain an active defense posture while aligning your efforts to real adversary behavior.
- **Test or Learn:**
 - Evaluate the EDR/SIEM effectiveness and visibility, or use EDRmetry for Red vs Blue team skill development in the next-generation, hands-on format of internal Linux security workshops/training

The updated list of supported offensive tests and additional information about EDRmetry Linux Matrix can be found on the project website:

- <https://edu.defensive-security.com/edrmetry-linux-matrix-for-download-advanced-hands-on-attack-ttps-catalog>

III. LINUX ATTACK, DETECTION, and FORENSICS v2.0 - HANDS-ON PURPLE TEAMING PLAYBOOK



Attackers continuously find new ways to attack and infect Linux systems, employing increasingly sophisticated techniques and tools. As defenders, we must stay current with adversaries, understand their techniques, tactics, and procedures (TTPs), and be able to respond rapidly. Combining low-level visibility across the network, servers, and endpoints is key to achieving this goal. When digital forensics and

incident response (DFIR) needs arise, we can go even further by applying proactive investigation of attacker activity traces. This course covers various use cases related to conducting and detecting attacks, analysis, artifact inspection, and response, while also teaching the essential aspects of effectively handling incidents on Linux systems.

The **"Linux Attack, Detection and Forensics v2.0 - Hands-on Purple Teaming Playbook"** course helps build and understand low-level attack paths against Linux systems, increases attack detection coverage, presents numerous open-source projects from the DFIR/defense domain, and explains the need for collecting telemetry data from Linux systems. The attack techniques and paths covered in this course include a wide variety of malware/rootkit implementations such as eBPF, XDP, io_uring, Ftrace, Kprobe, Uprobe, Netfilter, Systemtap, PAM, SSHD, HTTPD/Nginx, LD_PRELOAD-based code examples, and many others. The detection and analysis layers covered include LKRG, bpftool, Velociraptor IR, OSQuery, command-line level analysis of /proc/ and /sys/ locations, memory analysis using the Volatility 2/3 framework with partially automated RAM acquisition, Sysmon4Linux, Falco Runtime Security, Kunai Runtime Security, Tracee, Sysdig, Tetragon Runtime Security, BPF-LSM, Sandfly Security, Splunk, Elastic Security, Wazuh, Zeek IDS, Suricata IDS, Moloch/Arkime FPC, Yara rules, and many others.

The hands-on content is divided into sections mapped to the MITRE ATT&CK Framework. Through the dynamic offensive layer approach leveraging EDRmetry Linux Matrix, we present a fresh, unique, and above all, forward-looking approach to actively acquiring new knowledge and experience in the field of Linux active defense, so-called "Purple Teaming". If you want to deepen your understanding of the internal architecture of Linux x86/x64 systems and be prepared for Linux-related threats, this material is essential.

The extensive course agenda covers:

- Introduction
 - Welcome to the v2.0 party!
 - Goals / What to expect
 - Active Defense, PT, DE & Assume Breach
 - Linux DFIR Introduction

- PurpleLabs Detection and Hunting Dashboard
 - General Course Flow Design
 - Private Virtual Machines
 - Shared Virtual Machines
 - PurpleLabs VM Robot Tool
 - PurpleLabs Network Diagram
 - EDRmetry Matrix
 - EDRmetry Matrix + HexStrike AI Integration
 - Why Linux as a target?
 - Linux Threat Landscape
 - Open Source Community
 - Recommended books
 - [Changelog / Updates]
- Golden Rules - Before You Start Hands-on
 - Explore widely, think broadly
 - Enable Runtime Security/DFIR Agents
 - SOCKS Proxy or SSH Tunneling is required
 - KALI_X or C2_X?
 - TARGET_X - Kernel upgrade/downgrade
 - Multi-tab terminal
 - What is O- ?
 - "Don't" Policies
 - Troubleshooting
- Defensive/DFIR Tooling
 - Host/Exploration of CLI tools
 - Host/Exploration of /proc
 - Host/Exploration of /sys
 - Host/Splunk Forwarder
 - Host/Falco Runtime Security
 - Host/Kunai Runtime Security
 - Host/Tetragon Runtime Security
 - Host/Tetragon Custom Tracing Policies
 - Host/Tracee Runtime Security
 - Host/Jibril Runtime Security
 - Host/Elastic Security Agent

- Host/Wazuh Agent
 - Host/Velociraptor Agent
 - Host/FleetDM OSquery Agent
 - Host/Sandfly Security
 - Host/Sysmon4Linux
 - Host/Syslog
 - Host/auditd
 - Host/Linux IR Scripts - SecureProbes
 - Host/UAC
 - Host/Ghostscan
 - Host/Decloaker
 - Host/bpftrace
 - Host/LKRG
 - Host/SELinux
 - Host/Capa
 - Host/Yara / Yara-x Scanning
 - Network/Zeek NIDS
 - Network/Suricata NIDS
 - Network/WAF Modsecurity
 - Memory/Volatility3 Framework
 - Memory/Volatility2 Framework
 - Commercial Linux EDR/Security Products
- Linux EDR Architecture
 - Introduction
 - What is Linux EDR engine?
 - How does Linux EDR work?
 - Core functionalities and key features
 - Visibility Events / Indexes / Data sources
 - Syscalls, Kernel Functions and Tracing Visibility
 - Detection logic / rulesets
 - Support for Sigma Rules
 - Engine Modes
 - Dashboards, Analytics & Query language
 - Response, Triage and Forensics
 - Deployment and Operations
 - Alerts / Incidents / Detections

- Query Language
 - Linux EDR Telemetry Project
- Linux MITRE ATT&CK Matrix
 - Introduction
- Initial Access - TA0001
 - TA0001 - Introduction
 - EDR-T6261 - Remote UAF Exploitation - user
 - EDR-T6119 - Remote UAF Exploitation - root
 - EDR-T6354 - Remote UAF+Heap Overflow
 - EDR-T6062 - Kafka CVE-2023-25194
 - EDR-T6355 - Langflow API CVE-2025-3248
 - EDR-T6114 - ActiveMQ CVE-2023-46604
 - EDR-T6105 - Apache HTTP CVE-2021-41773
 - O-EDR-T6116 - Tomcat Upload Manager
 - EDR-T6110 - Solr Log4J JNDI CVE-2021-44228
 - EDR-T6233 - XZ Backdoor CVE-2024-3094
 - EDR-T6113 - Spring CVE-2022-22963
 - EDR-T6416 - React2shell - CVE-2025-55182
- Execution - TA0002
 - TA0002 - Introduction
 - EDR-T6277 - Built-in System Tools Execution
 - EDR-T6041 - Execute binary listening from a hidden directory
- Persistence - TA0003
 - TA0003 - Introduction
 - EDR-T6395 - LKM LibZeroEvil r00tme
 - EDR-T6394 - LKM Singularity Rootkit
 - EDR-T6100 - LKM Char Device + LPE
 - EDR-T6163 - LKM Reveng Rootkit
 - EDR-T6023 - LKM Diamorphine Rootkit
 - EDR-T6289 - LKM Ftrace Rootkit - Rebellion
 - EDR-T6161 - LKM BDS Ftrace Hooking Rootkit
 - EDR-T6154 - LKM Suterusu Rootkit
 - EDR-T6155 - LKM KoviD Rootkit

- EDR-T6152 - eBPF Boopkit Rootkit
 - EDR-T6151 - eBPF TripleCross Rootkit
 - EDR-T6104 - SSHD Dummy Cipher Suite BYOT
 - EDR-T6235 - LD_PRELOAD Re-adding SSH key
 - O-EDR-T6327 - LD_PRELOAD Father Rootkit
 - EDR-T6422 - bdvl - Patch Dynamic Linker
 - EDR-T6423 - SSHD id_ed25519 Key Backdoor
 - EDR-T6170 - Cap_setuid over LD linker
 - EDR-T6139 - Python .pth Extensions
 - O-EDR-T6407 - Shadow SUID binfmt_misc
 - EDR-T6093 - Crontab root Backdoor
 - EDR-T6431 - Supervisor persistence
 - EDR-T6015 - Systemd Backdoor service
 - EDR-T6179 - Udev+atd C2 persistence
 - O-EDR-T6213 - Malicious RPM package
 - EDR-T6164 - PAM Sneaky Backdoor
 - EDR-T6144 - DNF Package Manager
 - EDR-T6051 - Modify core_pattern
 - EDR-T6347 - Nginx Shell Module
 - O-EDR-T6017 - HTTPD mod_backdoor
 - EDR-T6011 - PHP Webshells
- Privilege Escalation - TA0004
 - TA0004 - Introduction
 - EDR-T6109 - LPE Socket Command Injection
 - EDR-T6360 - UAF Dirty Pagetable Huge pages Kernel LPE
 - EDR-T6359 - UAF Cross-cache Dirty Pipe LPE
 - EDR-T6184 - PATH Hijacking
 - O-EDR-T6315 - Add SSH key via iptables-save
 - EDR-T6335 - sudo chroot CVE-2025-32463
 - EDR-T6147 - Docker Escape with socket+Chisel
 - EDR-T6417 - Docker Escape with core_pattern
 - EDR-T6187 - NFS SUID Escalation
 - EDR-T6230 - pkexec CVE-2021-4034
 - EDR-T6231 - DirtyPipe CVE-2022-0847 LPE
 - O-EDR-T6177 - MySQL UDF Command Exec
 - EDR-T6460 - Dirty Frag - LPE

- Defense Evasion - TA0005
 - TA0005 - Introduction
 - EDR-T6039 - File Transfer to a hidden directory
 - EDR-T6138 - Bash HTTP GET data with /dev/tcp
 - EDR-T6173 - Hackshell + OpenSSL download
 - EDR-T6340 - Python HTTP POST and Exec
 - O-EDR-T6363 - Base64 Payload inside ZIP
 - EDR-T6370 - Sneaky_remap + Ptrace() Process Injection in Rust + SSL/TLS callback
 - EDR-T6439 - Malasada .so to Shellcode Loader
 - EDR-T6127 - dd+/proc/PID/mem Injection
 - EDR-T6256 - STOP/CONT Process Injection
 - EDR-T6108 - ASM Injection over /proc/PID/mem
 - EDR-T6037 - Python3 Fileless memfd_create
 - EDR-T6188 - Fileless Execution with memexec
 - EDR-T6171 - Easy Proc Name Masquerading
 - EDR-T6038 - Proc Name Masq with exec
 - EDR-T6140 - Proc Name Masq with prctl()
 - EDR-T6345 - Proc Masq with mount NS
 - EDR-T6046 - LD_PRELOAD Proc Hiding
 - EDR-T6053 - mount --bind process hiding
 - EDR-T6253 - eBPF Process Hider
 - EDR-T6107 - LKM Fileless Remote Loading
 - EDR-T6396 - LKM Hooking init_module
 - EDR-T6293 - LKM Disabling SELinux
 - EDR-T6282 - LKM Reset Yama ptrace_scope
 - EDR-T6167 - BOF Loading with BOF-Stager
 - EDR-T6067 - SSH notty session
 - EDR-T6133 - File immutable with chattr
 - EDR-T6089 - Bashrc File Hiding with ls Alias
 - O-EDR-T6078 - Execute tools via PRoot BYOF
- Command and Control - TA0011
 - TA0011 - Introduction
 - EDR-T6123.023 - curlrevshell
 - EDR-T6123.008 - Revshell openssl+/dev/fd/3

- EDR-T6123.004 - Revshell mkfifo+nc
- EDR-T6123.022 - Oneshell - echo and chmod
- EDR-T6123.015 - Revshell Python TLS
- O-EDR-T6126.002 - Sliver C2 MTLS
- EDR-T6126.013 - Sliver C2 HTTPS
- O-EDR-T6126.007 - Sliver C2 DNS
- O-EDR-T6126.011 - Sliver C2 TCP Pivots
- EDR-T6212 - Emp3r0r HTTP2 AES Stager C2
- EDR-T6317 - SOA/ECS DNS C2 Channel
- EDR-T6115 - DNS Tunneling with dnscat2
- O-EDR-T6409 - Venom C2
- O-EDR-T6126.006 - Platypus C2
- EDR-T6126.004 - Merlin HTTP2 C2
- EDR-T6126.010 - Mythic C2 Deployment
- O-EDR-T6126.003 - Mythic C2 Thanatos ELF
- O-EDR-T6126.009 - Mythic C2 Medusa Python
- O-EDR-T6126.005 - Mythic C2 Poseidon ELF
- O-EDR-T6148 - XOR shell_reverse_tcp Loader
- EDR-T6117 - UPX Reverse SSH server
- EDR-T6123.017 - Shell over HTTP streams
- EDR-T6328 - io_uring Agent

- Credential Access - TA0006
 - TA0006 - Introduction
 - EDR-T6242 - eBPF Spy on PAM with Python3
 - EDR-T6199 - eBPF pampspy
 - EDR-T6012 - Sniff sshd with strace
 - EDR-T6415 - ptrace() ssh-inject
 - EDR-T6319 - Dump heap memory from Java

- Discovery - TA0007
 - TA0007 - Introduction
 - EDR-T6225 - Execute "What Server"
 - EDR-T6040 - Execute LinPEAS from /dev/tcp
 - EDR-T6065 - /proc/PID/ Enumeration
 - EDR-T6251 - Process Snooping with pspy
 - EDR-T6280 - Find loaded eBPF programs/maps

- EDR-T6218 - Linux VM Check via Hardware
 - EDR-T6204 - Read local file using curl
 - EDR-T6010 - Check my public IP
 - EDR-T6412 - DNS Reconnaissance
- Lateral Movement - TA0008
 - TA0008 - Introduction
 - EDR-T6189 - Reverse SOCKS5 proxy
 - EDR-T6357 - Chisel Reverse Socks Proxy
 - EDR-T6255 - KCP - FRP Fast Reverse Proxy
 - EDR-T6033 - SSH Linux Tunneling
 - EDR-T6404 - mTLS Reverse SOCKS5
 - O-EDR-T6387 - SOCKS5 over Tailscale
 - EDR-T6392 - Cloudflared Tunneling
 - EDR-T6131 - Hijacking SSH Client Session
 - O-EDR-T6364 - Hijacking the SSH-Agent session
 - EDR-T6057 - Execute Port Scanning
- Exfiltration - TA0010
 - TA0010 - Introduction
 - EDR-T6120 - Python FTP Upload
 - EDR-T6180 - SMB Data Exfiltration
 - EDR-T6112 - NTP Data Exfiltration
 - EDR-T6137 - HTTP PUT method + transfer.sh
 - EDR-T6181 - Upload/download data over SSHFS
 - O-EDR-T6135 - Upload data over WebDAV
 - EDR-T6234 - pam_exec SSHD Exfiltration
 - O-EDR-T6103 - PAM creds over HTTP Post
 - O-EDR-T6168 - ICMP_exfil + nping
 - EDR-T6418 - LDAP Data Hiding - FreeIPA
- Impact - TA0040
 - TA0040 - Introduction
 - O-EDR-T6063 - Ransomware C - lokpack
 - O-EDR-T6252 - Crypto Mining CPU stress
 - O-EDR-T6018 - Ransomware bash+openssl

- Attack Flows & Combos
 - Introduction
 - Combo Flows
 - Attack Flow #1
 - Attack Flow #2
 - Attack Flow #3
- Linux Hardening
 - Linux Hardening Guide
 - Container Security Hardening
- Kubernetes
 - K8S Components
 - K8S Deployment
 - K8S gVisor Deployment
 - Falco Deployment in K8S
 - Tracee Deployment in K8S
 - Kubernetes Threat Matrix
 - Kubectl Fu
 - K8S Simple Pod Deployment
 - K8S Unauthenticated API Access
 - K8S Static Hidden Pod
 - K8S Pod Breakout
 - K8S Mounted secrets
 - K8S Etcd Dump
 - K8S CronJob SSH Persistence
 - K8S Pod Name Similarity
 - K8S Grafana + PostgreSQL RCE
 - K8S Vuln Web App
 - K8S Security Scanners
- Active Security Research
 - Extra Research
 - Linux Internals
 - LSM Sandboxes
 - eBPF
 - Exploitation

- Kubernetes
- Evasion / Bypassing techniques
- Rootkits
- Malware
- OS Security Stacks
- Memory Forensics
- Fun

The complete and always up-to-date agenda is available on the provider's website:

- <https://edu.defensive-security.com/linux-attack-detection-and-live-forensics-v2-0>

Lab Preview sections are available here:

- https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3343257-exfiltration-ta0010/10996308-edr-t6234-pam_exec-sshd-exfiltration
- <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3342720-lateral-movement-ta0008/11041962-edr-t6404-mtls-reverse-socks5>
- <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3343261-discovery-ta0007/11240626-edr-t6040-execut-e-linpeas-from-dev-tcp>
- <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3288961-defense-evasion-ta0005/11734278-edr-t6455-fake-tls-clienthello-dpi-bypass>
- https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3288961-defense-evasion-ta0005/10818257-edr-t6370-sneaky_remap-pttrace-process-injection-in-rust-ssl-tls-callback
- <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3342719-privilege-escalation-ta0004/11037866-edr-t63>

35-sudo-chroot-cve-2025-32463

- <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3331658-persistence-ta0003/11116998-edr-t6161-lkm-bds-ftrace-hooking-rootkit>
 - <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3350642-defensive-dfir-tooling/10922528-host-falco-runtime-security>
 - <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3350673-introduction/11353301-purplelabs-network-diagram>
 - <https://edu.defensive-security.com/view/courses/linux-attack-detection-and-live-forensics-v2-0/3350673-introduction/10960544-general-course-flow-design>
-

IV. PURPLE TEAMING EXERCISES / DETECTION ENGINEERING:

Purple teaming is a collaborative approach that combines elements of red teaming (attack simulation) and blue teaming (defense simulation). The red team works closely with the blue team (internal security defenders) to share insights and improve the overall security posture. The main value of the service is to increase detection coverage, identify detection gaps, and shorten response times.

Adversary emulation involves simulating the tactics, techniques, and procedures (TTPs) of real-world threat actors to assess an organization's security defenses and incident response capabilities. This process aims to replicate the behaviors of actual adversaries, allowing security teams to better understand their vulnerabilities and weaknesses. This proactive approach helps organizations enhance their security posture by identifying and addressing potential gaps in their defenses.

Adversary emulation is not a one-time activity; it's an ongoing process. Organizations can use the insights gained from these simulations to continuously

improve their security controls, detection capabilities, and incident response procedures.

Adversary emulation helps evaluate the effectiveness of security controls, including intrusion detection systems, firewalls, endpoint protection, and other defensive measures. This assessment is critical for identifying and addressing weaknesses in the security infrastructure.

Emulating real threat actors provides a realistic context for assessing security defenses. This approach goes beyond simple vulnerability scanning by considering how attackers might leverage multiple vulnerabilities and tactics in a coordinated manner.

Security teams create custom attack paths that replicate the steps a real attacker might take to compromise systems and achieve their objectives. These attack paths often involve a series of simulated steps, from initial access to data exfiltration.

Our offer includes:

- Emulating Real Threat Actors
- Custom Attack Paths
- Red Team Exercises
- Scenario-based Simulations
- Detection Engineering / Continuous Improvement
- Purple Teaming
- Evaluation of Security Controls

Learn more from our article describing the completed project - Linux Purple Team Exercises - Use Case Story for Deutsche Bahn:

- <https://www.defensive-security.com/blog/linux-purple-team-exercises-use-case-story>
 - <https://www.defensive-security.com/services/purple-teaming>
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V. POTENTIAL WAYS TO ESTABLISH COLLABORATION:

There are several potential ways to establish collaboration:

1. Individual (per user) access to the PurpleLabs environment (90 days+) + unlimited-time access to the "Linux Attack, Detection and Forensics v2.0 - Hands-on Purple Teaming Playbook" course materials + access to EDRmetry Linux Matrix + optionally periodic knowledge transfer in an online format, e.g., a 4-hour meeting once a month for a year to discuss and work through individual labs.
2. Organization of dedicated live trainings (3 to 5 days) at the client's office or online, in groups of 10-12 participants each.
3. Consulting services per MD:
 - Attack Path Emulations
 - Purple Teaming Exercises
 - Evaluation testing of SIEM/EDR visibility.
 - Linux Detection Engineering.
 - Linux Incident Response.
 - Linux Hardening.
 - Building and deploying a custom prevention policy / EDR Linux Agent.

VI. ABOUT DEFENSIVE SECURITY:

Defensive Security is a company with experience in advanced cybersecurity solutions and services, focused primarily on implementing and recommending proven, stable open-source solutions, securing Linux environments, and conducting active security testing of critical infrastructure, including attack simulations and evaluation of detection coverage. Our goal is to deliver professional services and deployments alongside active validation of detection and incident response effectiveness, directly in an offensive context.

Defensive Security provides technical knowledge-transfer services, live workshops and training, and online courses for self-paced learning and exploration. The

hands-on substantive content of the workshop materials covers active threat hunting within Linux infrastructure from a "Red vs Blue" - attack vs defense perspective. Strong emphasis is also placed on the analysis of low-level Linux components and subsystems, which in turn enables the acquisition of knowledge in the area of robust system hardening and default blocking of execution of components commonly leveraged during exploitation.

Defensive Security collaborates with global market leaders in the areas of consulting, service hardening, and the deployment and tuning of Runtime Security and EDR solutions for Linux and Kubernetes environments. The company's offering targets the building and advancement of Purple Teaming competencies, which in turn translates into raising the capabilities of SOC/CSIRT teams in incident handling. Narrow specialization, an innovative perspective, and complementary technical and business competencies have enabled the successful delivery of numerous key projects across all sectors of the economy, both in Poland and globally.

CUSTOMERS:

- Microsoft, Nokia, ING Tech, Warta, Government of Dubai, PZU, PGNiG, PGE, Stack Overflow, DailyMotion, Alior Bank, Ministry of Finance Poland, MON, BGK, Millennium Bank, PKO BP, Gdańskie Centrum Informatyczne, Nazwa.pl, Rekord Systemy Informatyczne, IBS, AXA, Aviva, Cinkciarz.pl, Rockwell Automation, Esky, LPP, ARiMR, TUV, Polkomtel, Biatel, Integrated Solutions, Orange Polska, Fujitsu, KOWR, ARiMR, Agora, PERN, Gdańskie Centrum Informatyczne, Ministries of National Defense, Trend Micro, Vodafone, Aquasec, IBM, AmiViz, Netspi, Western Governors University, CrowdStrike, Simplicity Poland, Beyoncyber, StepStone, Collective Sense, Government of Singapore, CyberPRO, PepsiCo, Barclays, VISA, Paypal, Digital14, CERT Belgium, JPMorgan Chase, DSO National Laboratories Singapore, European Commission Cyber Division, DailyMotion, TUV, CommonSpirit Health, ExtraHop, DHL, Continental, Optiver, Abbott, MWR, TrustedSec, Maybank SG, Emirates, Google, mLeasing, Tiktok, Trustwave, Logitech, PKP Informatyka, WatchTowr, Vodafone, Ratels, Admiral Group, Grupa Azoty, Allegro, Cisco, Arista Networks, Sandia National Laboratories US, UK Defence Science and Technology Laboratory, Deutsche Bahn, Adyen,

Singapore Centre for Strategic Infocomm Technologies, E.ON, CommonSpirit Health, Adyen, KPN, Kaspersky, Palo Alto and more.

VII. LESZEK MIS PROFILE / EXPERIENCE:



Leszek Miś - CEO/Security Researcher at Defensive-Security.com, a company delivering cybersecurity services based on Open Source software, including Red Team adversary emulations, detection coverage testing for Blue Teams, EDR effectiveness validation, incident response/DFIR support, courses and advanced workshops, and high-quality knowledge transfer. Over 20 years of hands-on experience in the Red/Blue domain. Areas of interest include the development of multi-stage attack paths mapped to MITRE ATT&CK, multi-layered detection paths known as detection engineering, Linux/network-related ML feature extraction, Linux internals with particular focus on KS/eBPF rootkits, detection engineering, log analysis, memory analysis, threat hunting, and the exploration of new offensive techniques in Linux/Kubernetes relative to DFIR/detection/defense techniques.

- a. **Leszek Miś** – lead trainer and solutions architect at Defensive Security, holding authorized certifications, among which the following should be noted:
 - Offensive Security Certified Professional (OSCP)
 - Red Hat Certified Architect (RHCA)
 - Red Hat Certified Security Specialist (RHCSS)
 - Red Hat Certified Data Center Specialist (RHCDS)
 - Red Hat Certified Virtualization Administrator (RHCVA)
 - Red Hat Certified Engineer (RHCE)
 - Red Hat Certified Instructor (RHCI)

- Comptia Security+
- Splunk Certified Architect

b. Competencies:

- 2024-2026 - Linux EDR detection coverage validation, Linux EDR tuning, automation of the Linux offensive environment domain, Linux incident response support, knowledge transfer and technical training, attack emulations, Purple Teaming exercises.
- Between 2020 and 2024, he developed advanced attack paths in the Linux/Windows systems domain within Active Directory network environments and conducted detection coverage research incorporating open-source solutions. Some projects involved substantive support in incident response and post-breach analysis. He carried out SIEM deployments, EDR integrations and consultations, and broad-scope consulting in the area of low-level network security technologies and Linux/Windows systems.
- Between 2015 and 2018, he worked at the USA startup Collective Sense as VP, Head of Cybersecurity, where he was primarily engaged in network security research, APT campaign analysis, anomaly detection and behavioral analysis of the network environment, business and strategic analysis for the company, as well as the implementation of new technical functionalities into a product covering network security, operating system, and web application domains.
- Oversees projects and develops the company's service offering.
- A well-known and highly regarded Red Hat trainer and examiner, having delivered training and examinations across the Red Hat Certified Architect / RHCSS / RHCE certification path.
- Author of the Linux Attack, Detection and Forensics v2.0 - Hands-on Purple Teaming Playbook course - recognized and

appreciated worldwide by thousands of people.

- Trainer at numerous global conferences: Black Hat USA 2019/2024, Black Hat Asia 2024/2025/2026, OrangeCON NL 2025/2026, X33FCON PL 2025/2026, Brucon BE 2018/2022, Flocon USA, OWASP Appsec USA, Hack In The Box Dubai/Singapore/Abu Dhabi/Amsterdam 2019/2022/2023, Out Of The Box Jakarta 2026, Confidence PL, PLNOG, Open Source Day, Red Hat Roadshow, OWASP Poland Chapter, ISSA Infotrams.

c. LinkedIn Profile:

- <https://www.linkedin.com/in/crony/>

d. X Profile:

- <https://x.com/cr0nym>

Recent testimonials and recommendations from training and projects:

- **Deutsche Bahn Purple Teaming Exercises + X33FCON TRAINING:**
 - Last year, I attended x33fcon and took the training "Effective Linux EDR/XDR Evaluation Testing for Red and Blue Team" by him. The training was excellent — the number of attack techniques covered, each discussed with remarkable depth and detail, was impressive. The attack matrix, course material, and lab environment Leszek built for this are genuinely outstanding. The learnings on how open source and commercial EDR tools will help find even advanced attacks in your environment was impressive. **My colleagues and I, who attended, were so convinced by this training that we decided to bring Leszek in this year for a dedicated workshop/ PT exercises.** The goal: Challenge our detection capabilities and toolset, and improve them together with him. Leszek adapted his lab to our specific environment, and our defenses were put through their paces. We held our own — a good benchmark for where our Linux detection capabilities are at.

Together, we developed detections for highly sophisticated attack techniques, and both sides walked away having learned a lot. If you work in Linux security, the training and workshop are worth your time. The attack techniques are exceptionally well-documented and thought through - the kind of material you can directly translate into better detections and a stronger security posture for your entire organization.

- **Black Hat Asia 2026:**

- Very good lab environment
- This is the first time learning blue side skills, but I was so excited and fun to improve my own skills, thank you so much for your dedication!
- The main highlight of the course for me was the quality of the materials presented. They are very easy to follow and remain grounded in the course objectives. Moreover, the content is well thought out and can be explored beyond the duration of the course. I also enjoyed the style of Leszek's teaching as it is direct and interactive. I do wish that the course would be slightly longer (3 days instead of 2), so that more materials can be covered in-person. I would like to thank Leszek for preparing and sharing the materials and for his ongoing lectures!
- Having fun with the in-depth Linux internal telemetry. Learning a lot about what EDR should look for and how to find the critical artifacts.
- The content is very detailed with explanations of the steps. Overall, the training is very useful, especially on the offensive techniques and detection queries.
- Gave me a lot of new insights into the blue team side of work, especially on how EDRs are not the final goal and how they work better as data sources for defenders.
- Thanks, Leszek, I really enjoyed your course and learned new things. Your online materials are very comprehensive, and the PurpleLabs lab setup is great.
- Engaging and knowledgeable instructor
- The lessons were very well documented. Learned many things. Will recommend this course.

- The materials are clear and easy to follow. Leszek not only covered the materials but also demonstrated the thought process behind his approach for detection.
- Very good content, I learn a lot!
- The trainer demonstrated strong knowledge of the subject matter. The training content and materials were practical and applicable to real-world situations.
- Thank you for the excellent course. Your real-world experience and EDRmetry Linux Matrix Playbook were an eye-opener. I can tell that with open-source tooling, it is usually very powerful and sometimes more than commercial. Able to understand the importance of telemetry and context. Would love future iterations to include AI-Assisted telemetry workflows and detection.
- It was a very satisfying lecture. The quality of content was excellent, and it covered exactly what I was looking for. I believe it will be very helpful for my future career.

- **OrangeCON 2025:**

- Great course, excellent added value with ongoing access to course material as well.
- Great training, excellent, thank you!
- Great research, learned a lot! Respect for all that Leszek achieved.
- I liked the dynamic style of the course very much.
- Also, the presentation/live demo at the end of the course was really nice. I like the idea behind the whole EDRmetry/PurpleLabs project!
- Just awesome! Great material, well-prepared training!
- Such a nice, knowledgeable instructor! I'm so happy that I managed to attend this course! and definitely will be highly recommended!
- Beyond this training, it would even be interesting to see a certification. I am excited and would love to see the services and tooling Leszek will release soon. :-)
- The training was great, I learned a lot of new stuff, and it was very good to refresh my mindset to look more from a practical attacker's point of view instead of just reading up on single techniques. Thanks for a great 3 days. It was inspiring stuff.
- Great training, pace was fine. Even though I'm not super experienced in Linux architecture and couldn't understand everything, it was still

useful. Timing for the practical sessions was on point, not too long and not too short. Thanks for this awesome training! Best conference training I've had so far :)

- Leszek has a lot of knowledge about Linux forensics, threat hunting, and incident response. From an offensive perspective, the most useful parts of the training are getting familiar with the monitoring tools available and learning more about the internals of these tools (including the Linux internals). It gives me new inspiration to dive deeper into some topics, such as eBPF and some interesting syscalls. The lifelong access to the course material is going to be a great resource as well.
- **Head of Cyber Security @ PZU Group - the biggest insurance company in Poland:**
 - Every organization must constantly strengthen its defense capabilities, testing, and researching; that's obvious in cybersecurity. Many thanks to Leszek Miś, who did an excellent job with the workshop "Linux Incident Response". He performed a lot of the tests, attack emulations, and playbooks, so we have improved our skills related to Linux hunting and responding to incidents detected by EDR. His experience, commitment, and enthusiasm are top-notch, and we can sincerely recommend Leszek's work to others.

The full list of recommendations is available at the following address:

- <https://edu.defensive-security.com/about>